PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	Sekina Murad
Program/Batch	B082-Networkwalks
Date	13 November 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM3: Maltego W2-PM5 (Zenmap Scanning) W2-PM_FINAL
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network
Permission secured from client?	Yes
Phases covered	**Phase 1:** Reconnaissance & Footprinting **Phase 2:** Scanning & Network Discovery **Phase 3-5:** In Progress

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the networkwalks.com domain using multiple Kali Linux tools (W2-PM1), collecting email addresses with Maltego (W2-PM3) and scanning my own local network with Nmap/Zenmap (W2-PM5). Together they show how an attacker moves from gathering public information about a target to mapping live hosts on a network. It is the Week 2 part of my ongoing internship program at Networkwalks.
All commands were run in Kali Linux on my own laptop, connected through the wlan0 Wi-Fi interface. Every step below includes the exact command or tool used, the result I observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux	Operating system used for all reconnaissance and scanning activities
WHOIS	Find domain registration details (owner, dates, name servers).
whatweb	Fingerprint web technologies (server, CMS, plugins, IP).
nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers of the website.
wafw00f	Detect whether a Web Application Firewall protects the site.
dnsrecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV).
Maltego	OSINT graphing tool; harvested email addresses related to the domain using transforms.
ip / ip route	Identify the local IP address, subnet and default gateway.
Nmap / Zenmap	Ping scan (nmap -sn) of the local subnet to find live hosts, IPs and MAC addresses.

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance against the networkwalks.com domain using six Kali Linux tools: **WHOIS, WhatWeb, Nslookup, Curl, Wafw00f and DNSRecon**. Each tool was used to collect a different type of information about the target.

First, I used **WHOIS** to obtain publicly available domain registration information. The domain is registered with **GoDaddy.com, LLC**, was created on 2019-11-06 and expires on 2027-11-06. Its name servers are **NS6135.HOSTGATOR.COM** and **NS6136.HOSTGATOR.COM**, the domain has client delete/renew/transfer/update locks set, and DNSSEC is unsigned.

I then used **WhatWeb** to identify technologies used by the website. The results identified **WordPress 7.1.4** and **WP Download Manager 3.3.76** on an Apache server, along with jQuery 3.7.1, Google Tag Manager, a public contact email (info@networkwalks.com), an HttpOnly __wpdm_client cookie and uncommon headers such as x-nginx-cache and x-endurance-cache-level. It also showed the 301 redirect from HTTP to HTTPS.

Using **Nslookup**, I resolved the domain name to its IP address. The non-authoritative answer identified **192.232.216.135**, the same IP address reported by WhatWeb.

I used **Curl** with the -I option to inspect the HTTP response headers. The server returned HTTP/1.1 200 OK with Server: Apache, a Permissions-Policy, Referrer-Policy: no-referrer-when-downgrade, X-Endurance-Cache-Level, X-nginx-cache: WordPress, a secure HttpOnly cookie and the WordPress REST API endpoint /wp-json/.

Next, I used **Wafw00f** to determine whether a Web Application Firewall was protecting the website. After only 2 requests, the result identified **ModSecurity (SpiderLabs)**.

Finally, I used **DNSRecon** to enumerate DNS records. The results showed the SOA and NS records (ns6135 and ns6136.hostgator.com), the MX record mail.networkwalks.com (192.232.216.135), the A record, an SPF TXT record (v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all), a Google site-verification TXT record and 8 SRV records for _autodiscover._tcp pointing to cpanelemaildiscovery.cpanel.net. The DNSSEC query failed on all nameservers, which is consistent with WHOIS reporting DNSSEC as unsigned.

4.2 OSINT with Maltego

For the second activity, I used **Maltego Graph (Desktop) 4.13.0** to gather open-source intelligence about the same domain. After logging in through the browser, Maltego brought me back to the application and I accepted the default settings (Default System Browser, Privacy Mode: Normal).

In the Entity Palette I searched for **Domain**, dragged it onto the graph and renamed the entity to networkwalks.com. I then right-clicked the entity, filtered for email-related transforms and ran **To Emails @domain [Search Engine]**. The transform returned one email address entity, the public contact address that WhatWeb had also found. The Transform Output panel shows that each run used 10 Maltego credits and that 170 credits remained afterwards.

Why hackers need this information: harvested email addresses show which accounts can be targeted with phishing or social engineering. Combined with the WordPress details found by WhatWeb, an attacker can search for related vulnerabilities and launch technology-specific attacks, for example WordPress attacks against this website. We will study those attack examples in later sections.

About Maltego: it was created by Paterva in 2008 in Pretoria, South Africa. It comes in multiple editions: the Community Edition is free with limited features, while paid plans unlock more results, data and commercial use. It can also be installed and used on Kali Linux, and many other transforms can be tried one by one for learning purposes.

4.3 Network Scanning with Zenmap

For the final activity, I used **Nmap** (the engine behind **Zenmap**) to perform network discovery on my local network. The practical required me to identify my local IP address and subnet, discover live hosts, identify their IP and MAC addresses, and generate a network topology.

I first used the Linux ip -br addr and ip route commands to identify my local IP address and LAN subnet, and the default gateway. I then scanned the subnet with sudo nmap -sn 10.20.199.0/24, which performs the same ping scan as the Ping Scan profile in Zenmap, to identify active hosts.

My local network details were:

- Interface: wlan0 (default gateway [redacted])
- IP address: [redacted]
- LAN subnet: [redacted] (mask [redacted])
- MAC address: [redacted]:74

Nmap scanned 256 IP addresses in 10.67 seconds and reported **1 live host**: hk.local ([redacted]), which is my own Kali machine. The gateway did not answer the ping probes, which is common on Wi-Fi networks that filter ICMP or isolate clients. I entered these results in the Networkwalks Zenmap practice lab and scored **100%**.

After completing the scan, I created the network topology diagram (Internet → Wi-Fi Gateway [redacted] → Kali Linux [redacted]) and saved it in PDF format as required by the practical task.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified WordPress 7.1.4 and WP Download Manager 3.3.76	Attackers may use exposed technology/version information to identify software requiring further security review	● **Medium**
2	Server IP address identifiable	Nslookup resolved the domain to 192.232.216.135	Provides information about the network location of the web service	● **Low**
3	HTTP technical information exposed	Curl returned HTTP response headers and exposed /wp-json/	May assist technology fingerprinting and further enumeration	● **Low**
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs)	Reveals information about the web application's security architecture	● **Low**
5	DNS infrastructure information exposed	DNSRecon identified DNS, mail and service-related records	DNS information can help build a broader infrastructure profile	● **Medium**
6	Contact email address harvestable	Maltego and WhatWeb both exposed the public contact email info@networkwalks.com	Public addresses can be targeted with phishing and social engineering	● **Low**
7	Local host details visible on the LAN	Nmap and ip commands showed 10.20.199.181 and MAC 28:df:eb:0e:41:74	Device identifiers can be seen by other devices on the same network	● **Low**

Risk level key: ● Critical ● Medium ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities.

The practical exercises primarily involved information gathering, email harvesting and host discovery. No exploitation or vulnerability validation was performed as part of these modules.

Therefore, the presence of information such as a software version, IP address or DNS record does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

1. **Review publicly exposed technology information**
   Organizations should regularly review what information about their web technologies, CMS and plugins is publicly visible.
2. **Keep software updated**
   CMS platforms, plugins and other web technologies should be regularly updated and reviewed against current security advisories.
3. **Review HTTP headers**
   HTTP response headers should be reviewed to determine whether unnecessary technical information is being exposed.
4. **Review DNS records regularly**
   DNS records should be checked periodically to ensure that only required information and services are publicly exposed.
5. **Properly configure and monitor the WAF**
   Keep the WAF (ModSecurity) enabled and tuned, since it already blocks naive attacks.
6. **Perform regular internal network discovery**
   Organizations should periodically scan their own networks to identify active devices.
7. **Limit exposed email addresses**
   Use contact forms or role-based addresses so that harvested email addresses are less useful for phishing.
8. **Maintain network documentation**
   Network topology and device information should be documented and updated regularly.
9. **Perform security testing with authorization**
   Reconnaissance and scanning should only be performed against systems and networks where appropriate authorization has been provided.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, OSINT reconnaissance and network scanning.

In the footprinting activity, I used six Kali Linux tools to collect information about the target domain. I learned how WHOIS can provide domain information, WhatWeb can identify web technologies, Nslookup can resolve domain names, Curl can inspect HTTP headers, Wafw00f can identify a WAF, and DNSRecon can provide additional DNS information.

With Maltego I learned how a single domain entity and one transform can harvest email addresses, and in the network scanning activity I used Nmap/Zenmap to identify my local network configuration, discover active hosts and collect IP and MAC address information. I also created a network topology diagram.

The exercises showed me that information gathering is an important part of cybersecurity. Even before attempting to exploit a system, a security professional can learn a significant amount about an environment by carefully analyzing publicly available information and network responses.

I also learned that technical findings should be documented clearly. A good cybersecurity report should explain what was performed, what was discovered, what the observation means, what risk it may create, and what can be done to reduce that risk.

Finally, I learned that reconnaissance and scanning must always be performed within an authorized scope. These activities were completed as part of the assigned educational cybersecurity lab.

8. Evidences Collected

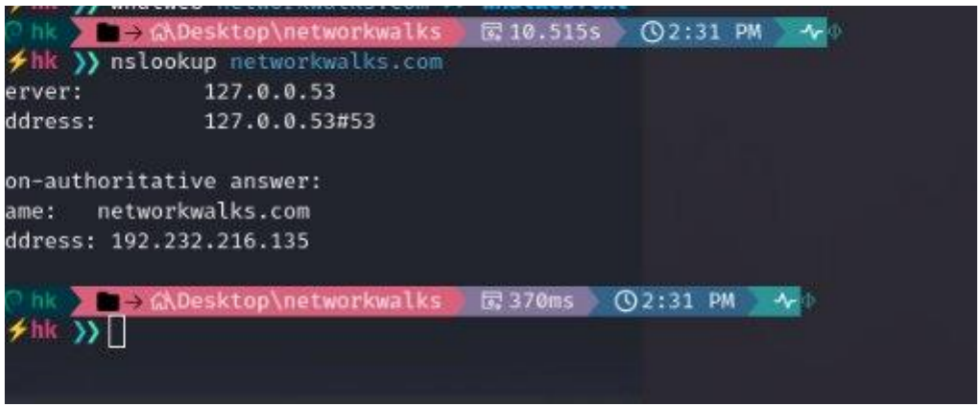
```
hk  5ms  2:11 PM
hk ❯❯ whois networkwalks.com
  Domain Name: NETWORKWALKS.COM
  Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
  Registrar WHOIS Server: whois.godaddy.com
  Registrar URL: http://www.godaddy.com
  Updated Date: 2025-11-12T10:08:43Z
  Creation Date: 2019-11-06T22:51:46Z
  Registry Expiry Date: 2027-11-06T22:51:46Z
  Registrar: GoDaddy.com, LLC
  Registrar IANA ID: 146
  Registrar Abuse Contact Email: abuse@godaddy.com
  Registrar Abuse Contact Phone: 480-624-2505
  Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
  Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
  Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
  Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
  Name Server: NS6135.HOSTGATOR.COM
  Name Server: NS6136.HOSTGATOR.COM
  DNSSEC: unsigned
  URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>> Last update of whois database: 2026-11-13T19:11:18Z <<<

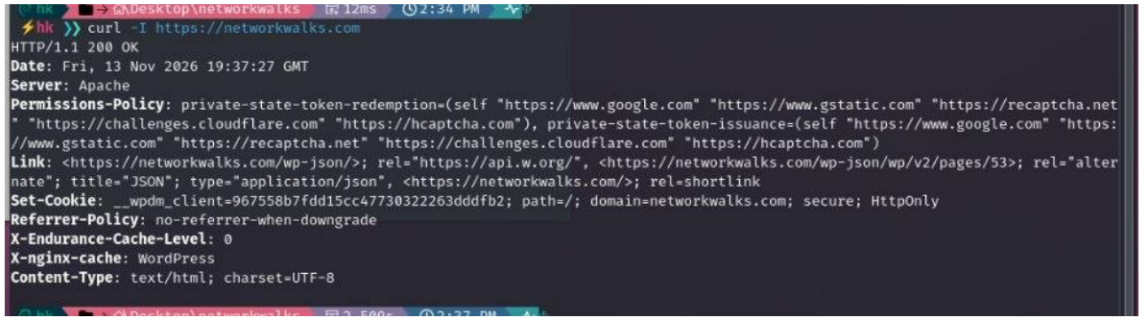
or more information on Whois status codes, please visit https://icann.org/epp

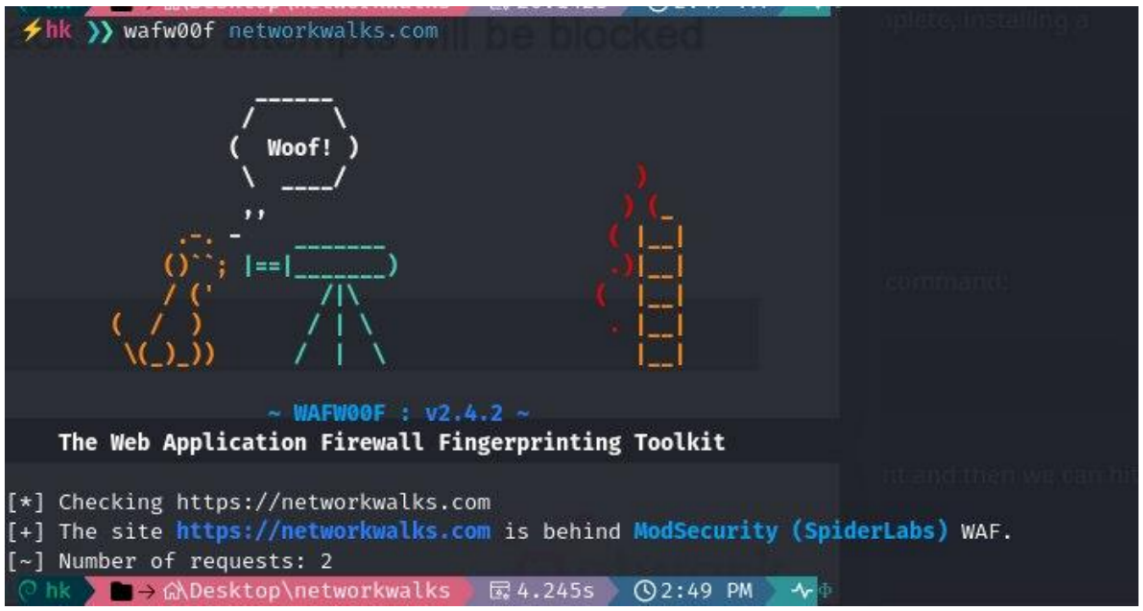
OTICE: The expiration date displayed in this record is the date the
egistrar's sponsorship of the domain name registration in the registry is
urrently set to expire. This date does not necessarily reflect the expiration
ate of the domain name registrant's agreement with the sponsoring
egistrar.  Users may consult the sponsoring registrar's Whois database to
iew the registrar's reported date of expiration for this registration.

ERMS OF USE: You are not authorized to access or query our Whois
atabase through the use of electronic processes that are high-volume and
utomated except as reasonably necessary to register domain names or
odify existing registrations; the Data in VeriSign Global Registry
ervices' ("VeriSign") Whois database is provided by VeriSign for
nformation purposes only, and to assist persons in obtaining information
bout or related to a domain name registration record. VeriSign does not
uarantee its accuracy. By submitting a Whois query, you agree to abide
y the following terms of use: You agree that you may use this Data only
or lawful purposes and that under no circumstances will you use this Data
o: (1) allow, enable, or otherwise support the transmission of mass
nsolicited, commercial advertising or solicitations via e-mail, telephone,
r facsimile; or (2) enable high volume, automated, electronic processes
hat apply to VeriSign (or its computer systems). The compilation,
epackaging, dissemination or other use of this Data is expressly
rohibited without the prior written consent of VeriSign. You agree not to
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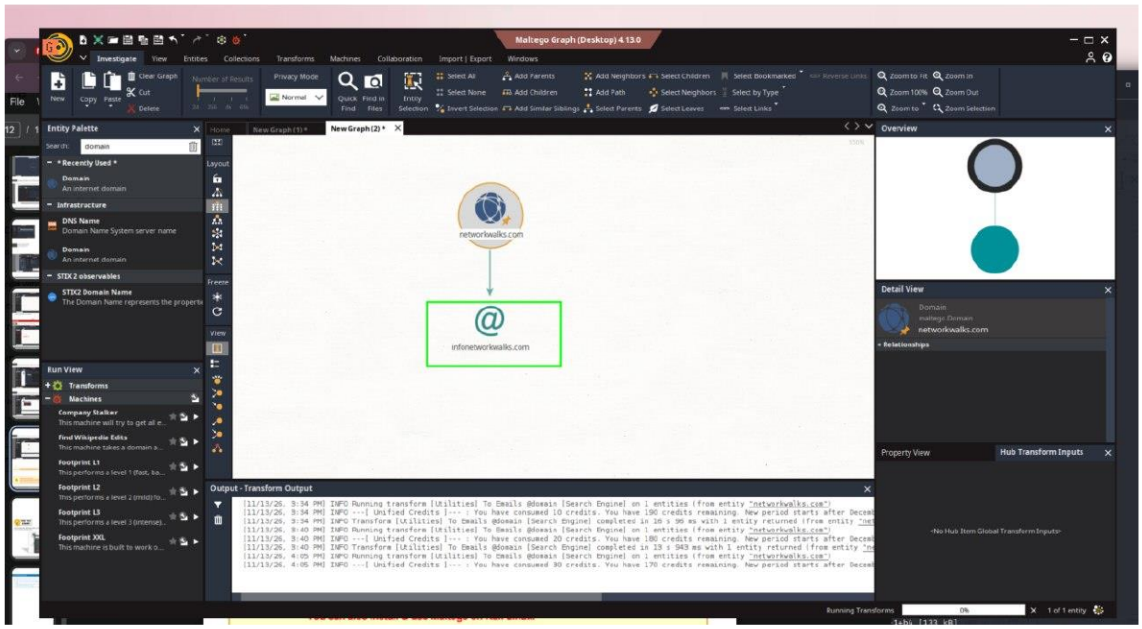
```
hk  ~\Desktop\networkwalks  4ms  2:18 PM
hk ❯❯ whatweb networkwalks.com
ttp://networkwalks.com [301 Moved Permanently] Apache, Cookies[__wpdm_client], Country[UNITED STATES][US], HTTP
erver[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/], Uncom
onHeaders[permissions-policy,x-redirect-by,referrer-policy,x-endurance-cache-level,x-nginx-cache]
ttps://networkwalks.com [200 OK] Apache, Bootstrap[7.1.4], Cookies[__wpdm_client], Country[UNITED STATES][US],
mail[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[__wpdm_client], IP[
92.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.4,WordPress Download Manager 3.3.76], Open-Graph-Pr
tocol[website], Script[4684NR-IPIB&amp;pidnVar2=50511&amp;prtVar2=7&amp;scvVar2=12,application/json,application
ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-poli
y,link,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.4]
ttps://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1.4], Country[UNITED STATES][US], Email[info@networkwalks
com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[W
rdPress 7.1.4,WordPress Download Manager 3.3.76], Open-Graph-Protocol[website], Script[4684NR-IPIB&amp;pidnVar2
50511&amp;prtVar2=7&amp;scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javascript
, Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,referrer-policy,x-endurance-cache-level,
-nginx-cache], WordPress[7.1.4]
hk  ~\Desktop\networkwalks  10.776s  2:19 PM
hk ❯❯
```


Desktop\networkwalks
10.515s
2:31 PM
hk ❯❯ nslookup networkwalks.com
erver: 127.0.0.53
ddress: 127.0.0.53#53
on-authoritative answer:
ame: networkwalks.com
ddress: 192.232.216.135
Desktop\networkwalks
370ms
2:31 PM
hk ❯❯


hk ❯❯ curl -I https://networkwalks.com
HTTP/1.1 200 OK
Date: Fri, 13 Nov 2026 19:37:27 GMT
Server: Apache
Permissions-Policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
Link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel=shortlink
Set-Cookie: __wpdm_client=967558b7fdd15cc47730322263dddfb2; path=/; domain=networkwalks.com; secure; HttpOnly
Referrer-Policy: no-referrer-when-downgrade
X-Endurance-Cache-Level: 0
X-nginx-cache: WordPress
Content-Type: text/html; charset=UTF-8


hk ❯❯ wafw00f networkwalks.com
Woof!
~ WAFW00F : v2.4.2 ~
The Web Application Firewall Fingerprinting Toolkit
[*] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[~] Number of requests: 2
Desktop\networkwalks
4.245s
2:49 PM

```
hk  ~\Desktop\networkwalks  1.407s  2:59 PM
hk ❯❯ dnsrecon -d networkwalks.com
2026-11-13T14:59:19.979164-0500 INFO Starting enumeration for domain: networkwalks.com
2026-11-13T14:59:19.979469-0500 INFO std: Performing General Enumeration against: networkwalks.com...
2026-11-13T14:59:20.355983-0500 ERROR All nameservers failed to answer the DNSSEC query for networkwalks.com
2026-11-13T14:59:20.842170-0500 INFO     SOA ns6135.hostgator.com 50.87.144.87
2026-11-13T14:59:21.312893-0500 INFO     NS ns6136.hostgator.com 192.232.216.131
2026-11-13T14:59:27.320861-0500 INFO     NS ns6135.hostgator.com 50.87.144.87
2026-11-13T14:59:34.252725-0500 INFO     MX mail.networkwalks.com 192.232.216.135
2026-11-13T14:59:34.257304-0500 INFO     A networkwalks.com 192.232.216.135
2026-11-13T14:59:35.585028-0500 INFO     TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
2026-11-13T14:59:35.585372-0500 INFO     TXT networkwalks.com google-site-verification=rr04eRmqHoWY3XemnizDNVK4q75X-Ij-mjgEeg-UsYI
2026-11-13T14:59:36.474035-0500 INFO Enumerating SRV Records
2026-11-13T14:59:38.656891-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.14 443
2026-11-13T14:59:38.657263-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.9 443
2026-11-13T14:59:38.657421-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.15 443
2026-11-13T14:59:38.657540-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.9 443
2026-11-13T14:59:38.657650-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.14 443
2026-11-13T14:59:38.657759-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.204.11 443
2026-11-13T14:59:38.657872-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.11 443
2026-11-13T14:59:38.657983-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.net 184.94.203.15 443
2026-11-13T14:59:38.711861-0500 INFO 8 Records Found
2026-11-13T14:59:38.712365-0500 INFO Completed enumeration for domain: networkwalks.com
```




```
hk  ~\Desktop\networkwalks  4ms  3:42 PM
hk ❯❯ ip -br addr
lo               UNKNOWN        127.0.0.1/8 ::1/128
wlan0            UP             10.20.199.181/24 fe80::2adf:ebff:fe0e:4174/64
docker0          DOWN           172.18.0.1/16
br-0d4ee04b7f3d  DOWN           172.19.0.1/16
hk  ~\Desktop\networkwalks  6ms  3:42 PM
hk ❯❯ ip route
default via [redacted] dev wlan0 proto dhcp src [redacted] metric 600
[redacted] dev wlan0 proto kernel scope link src [redacted] metric 600
[redacted] dev docker0 proto kernel scope link src [redacted] linkdown
[redacted] dev br-0d4ee04b7f3d proto kernel scope link src [redacted] linkdown
hk  ~\Desktop\networkwalks  4ms  3:42 PM
```

```
hk  ~\Desktop\networkwalks  4ms  3:51 PM
hk ❯❯ sudo nmap -sn 10.20.199.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-11-13 15:51 -0500
Nmap scan report for hk.local (10.20.199.181)
Host is up.
Nmap done: 256 IP addresses (1 host up) scanned in 10.67 seconds
hk  ~\Desktop\networkwalks  10.716s  3:51 PM
```

1. **Task1. Have you downloaded & installed Zenmap from official website?**

Yes

Not yet

2. **Task2. What is your local IP address & your LAN subnet?**

Network interface: wlan0
Local IP address [redacted]
Subnet mask: [redacted]5.255.0
LAN subnet [redacted]

3. **Task3. Copy & paste the list of live hosts/PC's in your IP subnet:**

Starting Nmap 7.99 (https://nmap.org) at 2026-11-13 15:51 -0500
Nmap scan report for hk.local [redacted]
Host is up.
Nmap done: 256 IP addresses (1 host up) scanned in 10.67 seconds

4. **Task4. How many hosts are live in your subnet?**

1 host was detected by Nmap.

5. **Task5. What are the IP addresses of the live hosts?**

10.20.199.181

6. **Task6. What are the MAC addresses of the live hosts?**

28:df:eb:0e:41:74

Zenmap Network Scanning (Practice Lab)

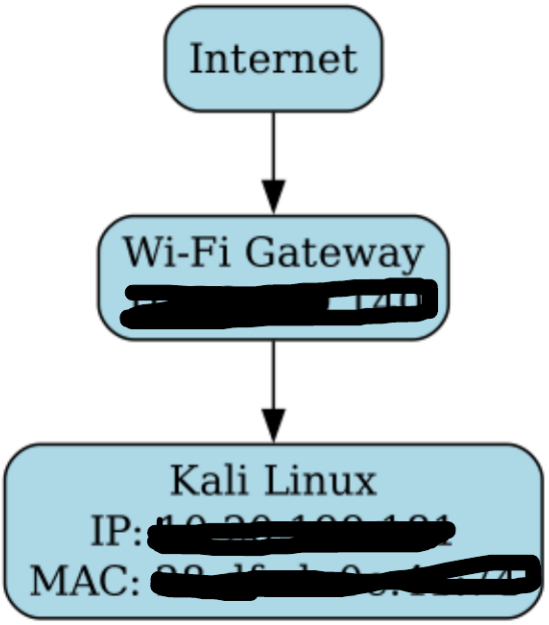
Please follow the PDF Lab guide to complete all tasks & answer below questions. Please use the lab files that have been shared with yo
by your Course admin (for Zenmap Network Scanning LAB).



Congratulations!

You have completed Zenmap Network Scanning (Practice Lab).

You have scored: **100%**



-End-

Author

Sekina **Murad**
Cybersecurity Student & Intern
GitHub: https://github.com/H-K32

Project Information

Program Name: Cybersecurity program at Networkwalks | **Week:** 02 | **Repository:** GitHub